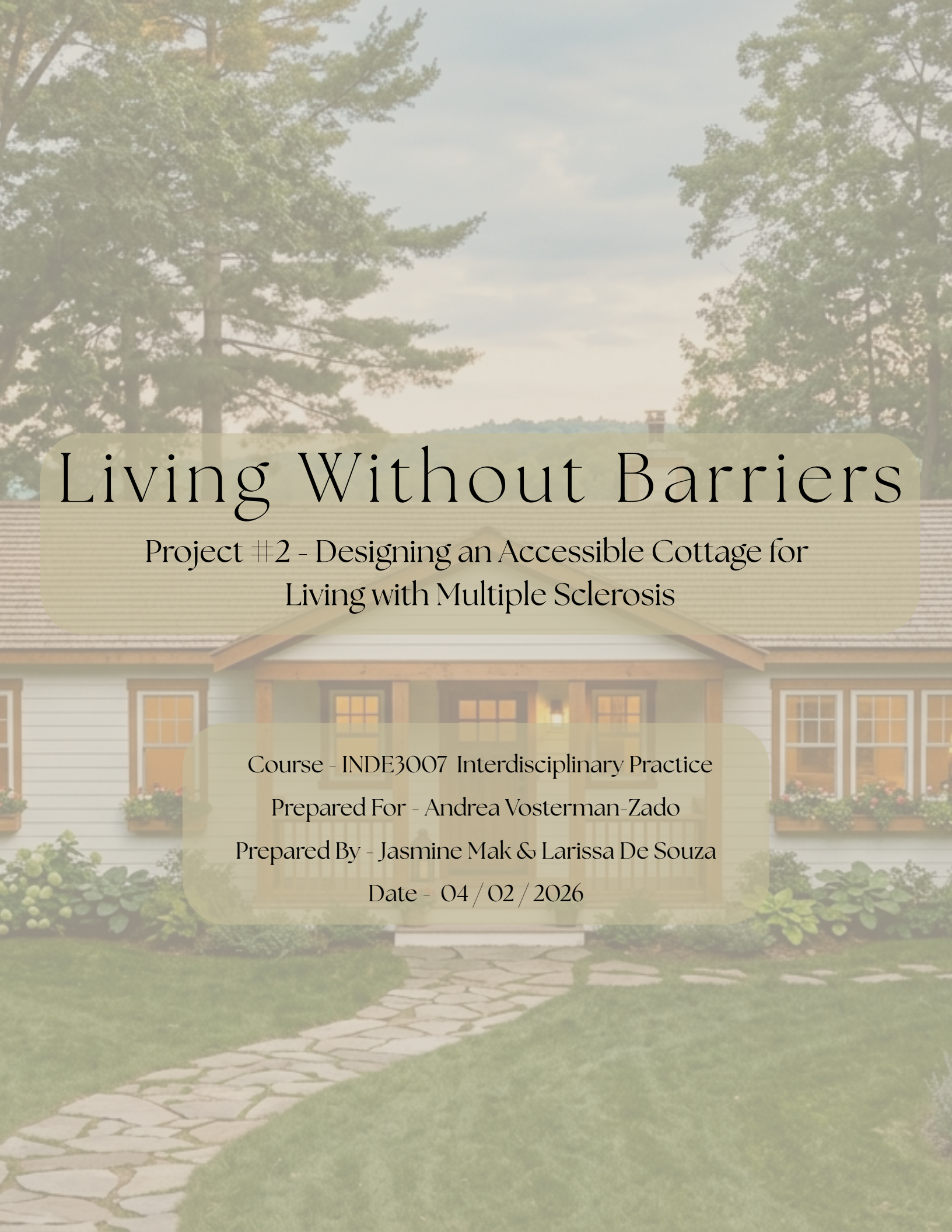


# Living Without Barriers

## Project #2 - Designing an Accessible Cottage for Living with Multiple Sclerosis

Course - INDE3007 Interdisciplinary Practice

Prepared For - Andrea Vosterman-Zado

Prepared By - Jasmine Mak & Larissa De Souza

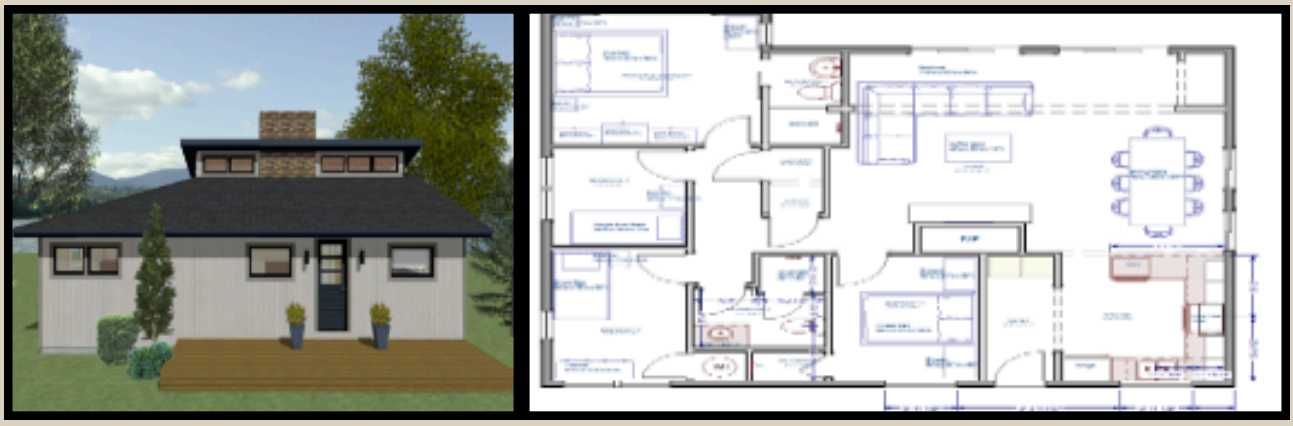
Date - 04 / 02 / 2026

# Abstract



***Living Without Barriers*** is a renovation project focused on redesigning an existing Muskoka cottage for a client living with ***Multiple Sclerosis***. The design prioritizes ***accessibility, independence, and safety*** through open circulation, accessible storage, and barrier-free kitchen and bathroom design. The project was developed through interdisciplinary collaboration with Occupational Therapy students, which informed accessibility requirements such as turning ***radius, reach ranges, and safe circulation paths***. The final design demonstrates how interior design can support ***independence, comfort, and quality of life*** through accessible and human-centered design strategies while maintaining a ***warm cottage environment*** connected to the surrounding landscape.

# Design — Site + Client Needs



The project involved the renovation of an existing single-storey cottage in Muskoka to better support a client living with Multiple Sclerosis. The original layout ***created barriers for mobility and accessibility***, so the design focused on ***improving circulation, accessibility, and safety*** throughout the home. Client needs and insights from the MS simulation and empathy mapping activities informed many of the design decisions.

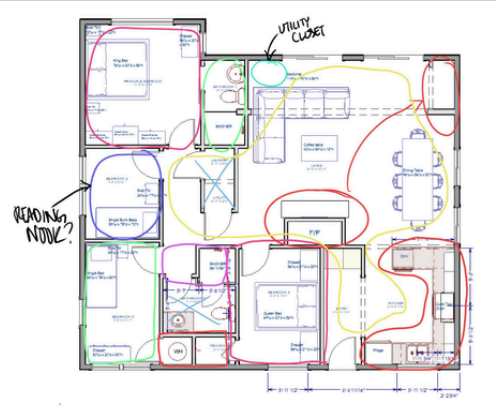
## Site Conditions

- Existing cottage renovation
- Single storey bungalow
- Original layout had small rooms and narrow circulation
- Limited accessibility for wheelchair use
- Renovation focused on open concept layout and improved circulation

## Client Needs

- Transitioning from walker to wheelchair
- Limited upper body reach
- Reduced grip strength
- Fatigue and mobility limitations
- Need for accessible bathroom and kitchen
- Need for safe circulation and rest areas

# Design — Design Development





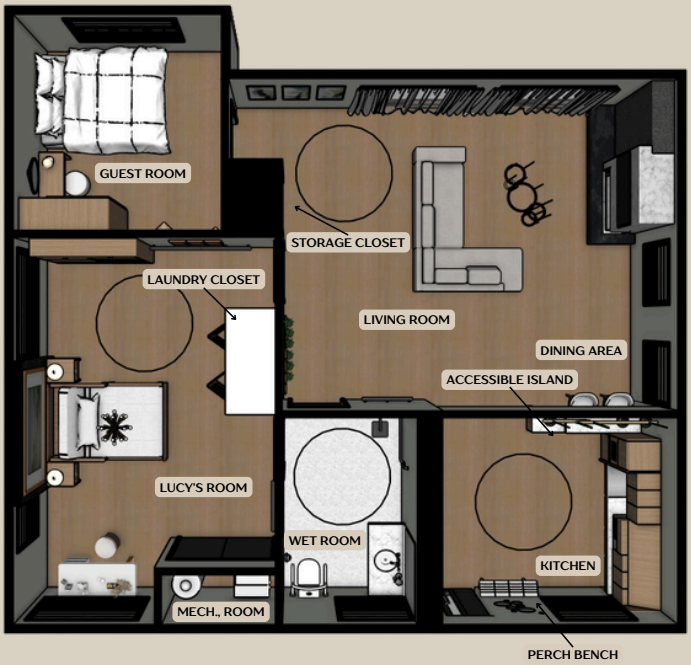





Multiple design options were explored to ***improve accessibility, circulation, and functionality*** within the cottage. Each layout considered ***wheelchair movement, accessible storage, bathroom layout, and kitchen accessibility***. Feedback from faculty and Occupational Therapy Assistant students helped refine the final design, which was selected based on circulation efficiency, accessibility, and convenience between key spaces.

- Explored multiple layout options
- Focus on wheelchair circulation
- Considered reach ranges and turning radius
- Improved bathroom accessibility
- Improved kitchen accessibility
- Feedback from OTA students
- Final design selected for best circulation and accessibility

# Design — Final Design Solution




The final design solution focuses on creating an ***open and accessible layout*** that supports ***independent living and safe movement*** throughout the cottage. The design reduces ***physical barriers, improves circulation***, and incorporates ***accessible kitchen and bathroom*** design features. Seating areas and clear movement paths were included to ***address fatigue and mobility challenges.***

## Final Design Strategy

- Open concept layout for easier circulation
- Wide hallways and turning radius
- Accessible kitchen with lower counter section
- Pull-out shelves and accessible storage
- Roll-in shower with grab bars and bench
- Non-slip flooring throughout
- Lever door handles
- Accessible laundry location
- Seating/rest areas throughout the home
- Smart home and voice-controlled systems

# Systems Coordination + WELL Design Integration

The design integrates building systems and WELL principles to create a safe, comfortable, and accessible environment that supports independence and long-term wellbeing.



## Building Systems Integration

**LIGHTING**

- Natural + layered lighting
- Wall lighting for circulation
- Voice-controlled lighting

**HVAC + AIR**

- Ventilation and air quality
- HVAC for temperature control

**ACCESSIBILITY**

- Accessible outlet locations
- Non-slip flooring
- Accessible appliances
- Barrier-free circulation

## WELL Standards

**LIGHT**

- Natural light
- Layered lighting

**AIR**

- Ventilation
- Indoor air quality

**COMFORT**

- Ergonomic furniture
- Accessible layout

**MIND**

- Connection to nature
- Calm cottage environment

# Conclusion

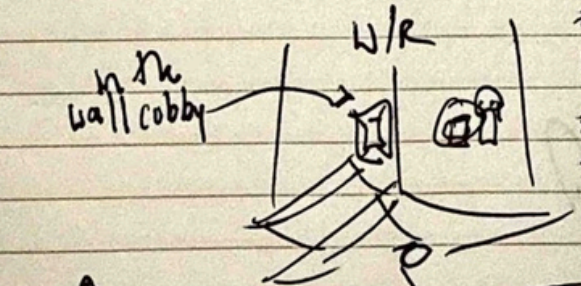


This project demonstrates how interior design and interdisciplinary collaboration can improve ***independence, safety, and quality of life*** for individuals living with Multiple Sclerosis. The final cottage design focuses on ***accessibility, open circulation, and safe kitchen and bathroom design*** while maintaining a comfortable residential environment. Collaboration with Occupational Therapy Assistant students helped inform accessibility requirements and improve the final design solution.

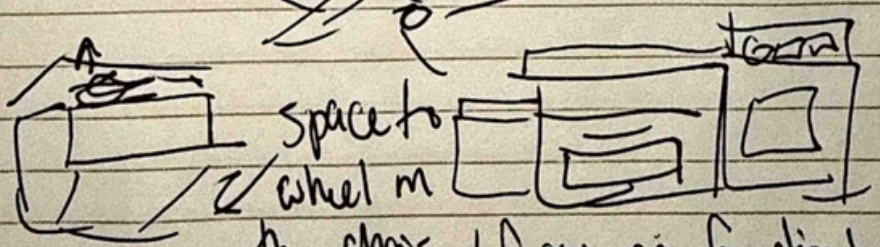
Friday March 6th / OT Meeting DATE: / /

Client IS Lucy! : 60yrs old

* wheelchair bound : in in a walker!
* upper body mobility issues - a lot of upper body strength needed
* fatigue, dizziness, bad balance
* living life in a wheelchair
  - ↳ range of motion to take of consideration
  - ↳ living life being sat
* need lots of space
* things need to be low - if it is too high might need to be raised
* laundry machine need to be side by side
* need to reach things
* outlet considerations
* put a lift in the bathroom
* BIG ROOMS! OPEN! the appliances lower
* the drawers to be in the middle



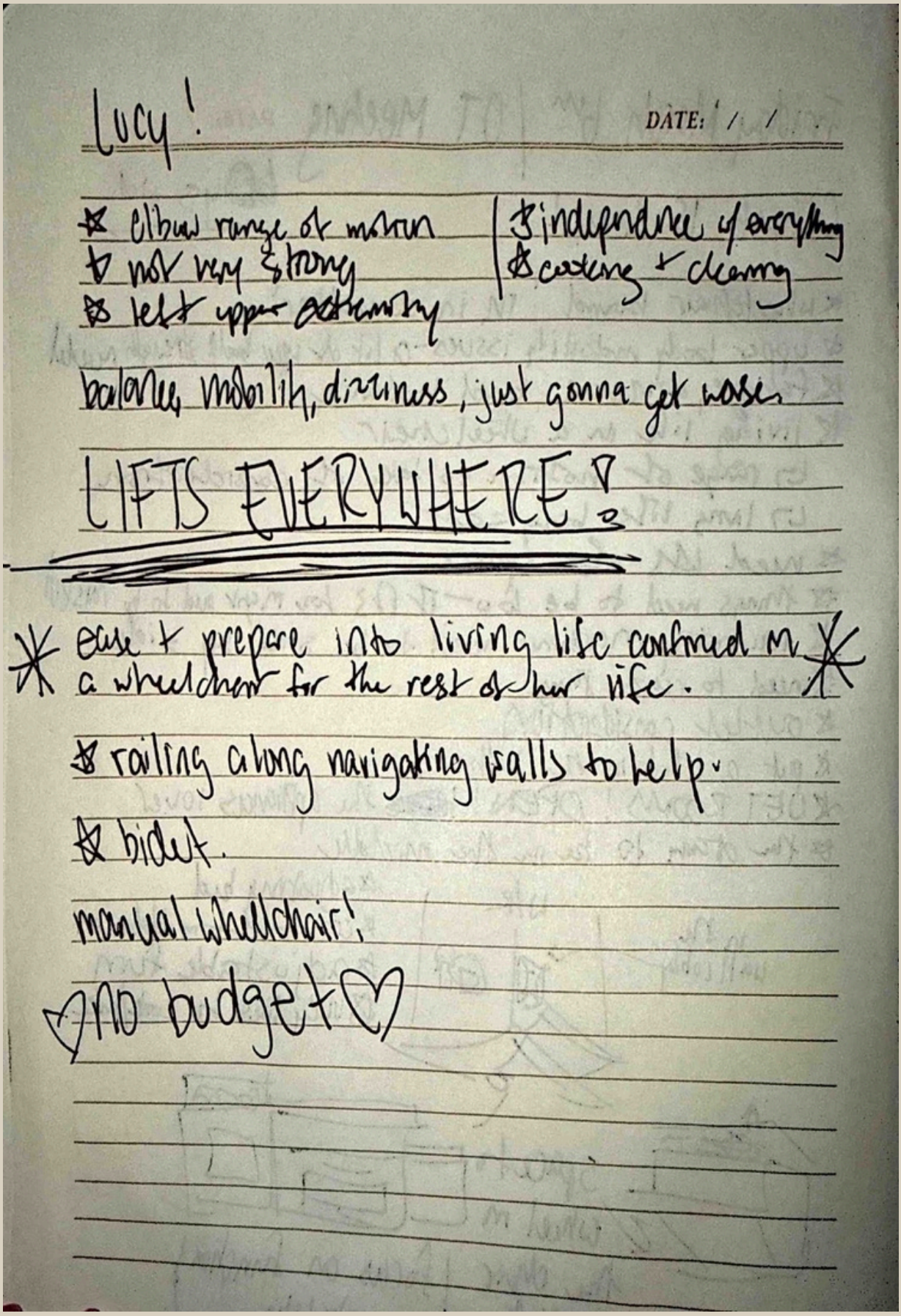

* adjusting bed
* crane + lift in bathroom
* adjustable turn
* Lucy likes to cook/bake



space to wheel in the chair underneath.

{focus on function + mobility}

# Appendix - 1st Milestone



Lucy!

DATE: / /

- elbow range of motion
- not very strong
- left upper extremity

- independence of everything
- cooking + cleaning

balance, mobility, dizziness, just gonna get worse

LIFTS EVERYWHERE!

- ease + prepare into living life confined in a wheelchair for the rest of her life.
- railing along navigating walls to help.
- bidet.

manual wheelchair!

♡no budget♡

# OT Meeting for MS

Project: Interdisciplinary Practice – MS Accessible Cottage

Client: Lucy

Location: M Building

Date: Friday, March 6th, 2026

Time: 2:00-3:30

Participants: Charlotte Sanderson, Ethan Garland, Kushi Patel, and Hui Lin (Sophia)

## Client Overview:

Lucy is recovering from a recent injury and has also been diagnosed with Multiple Sclerosis (MS). She currently experiences balance issues, fatigue, dizziness, and weakness, particularly in her left upper extremity. Lucy currently uses a rollator walker but may soon transition to a manual wheelchair as her mobility needs change.
Lucy would like to regain independence in daily activities such as meal preparation, laundry, and cleaning, and the design must support these goals while accommodating reduced mobility and strength.



**MOBILITY:**
- Uses Walker
- May transition to manual wheelchair



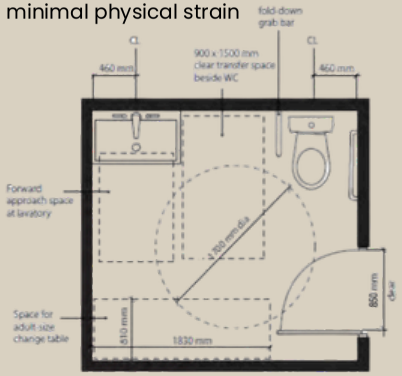
**PHYSICAL LIMITATIONS:**
- Limited elbow range of motion
- Weak grip strength
- Reduced upper body strength



**SYMPTOMS:**
- Fatigue
- Dizziness
- Balance Issues

## OT Insights

- Lucy's condition may progress, meaning the design must be flexible and adaptable
- Clear circulation is essential for wheelchair mobility
- Reduced strength and grip affects reaching, lifting, and handling objects
- Fatigue means daily tasks should require minimal physical strain




## Design Implications

- Future wheelchair circulation required (minimum 5ft turning radius)
- Storage and appliances must be reachable from seated position
- Reduce physical strain due to fatigue
- Design must adapt as MS progresses

## Design Opportunities

**Kitchen**
- Lower appliance placement
- Open layout circulation
- Knee clearance under counters
- Reachable storage

**Laundry**
- Side-by-side washer + dryer
- Front loading machines
- Clear floor space for wheelchair access

**Bathroom**
- Grab bars
- Accessible fixtures
- Safety support near toilet

**Bedroom**
- Adjustable bed
- Clear circulation around bed
- Accessible furniture layout

## Client Functional Priorities

- Maintain independence in daily activities
- Safe mobility throughout the home
- Reduce fatigue during everyday tasks
- Accessible kitchen for meal preparation
- Safe bathroom with fall prevention
- Efficient laundry access

Appendix - 1st Milestone

# Desiging for the Future

**Designing for a Transition to a Wheelchair**

Right now Lucy uses a rollator walker, but the OT mentioned she may transition to a manual wheelchair. So the space needs to work for both conditions.
Future planning considerations:

- Maintain wide circulation paths so a wheelchair can move easily through the apartment.
- Ensure doorways and hallways are wide enough for wheelchair access.
- Plan layouts that allow a turning radius in key areas like the kitchen, bathroom, and bedroom.
- Avoid tight corners or furniture that blocks pathways.

**Design implication:**
Even if she's using a walker today, designing circulation for a wheelchair now prevents costly changes later.

**Adaptable Storage and Reach Zones**

Lucy already has limited upper body strength and restricted elbow movement, and this could worsen over time.
Future planning considerations:

- Keep frequently used storage in lower, reachable areas.
- Avoid overhead cabinets for essential items.
- Use pull-out shelves or drawers instead of deep cabinets.
- Consider adjustable or modular shelving systems that can change height.

**Design implication:**
Storage should be usable from a seated position if Lucy eventually relies on a wheelchair.

**Safety and Stability Over Time**

Balance issues and dizziness increase the risk of falls, and these symptoms may become more severe.
Future planning considerations:

- Provide support elements such as grab bars or railings in critical areas.
- Ensure flooring is non-slip and stable.
- Avoid level changes or obstacles that could cause tripping.

**Design implication:**
The design should prioritize safe movement throughout the space.

**Flexible Furniture and Equipment**

Furniture should support Lucy now but also be adaptable later.
Examples:

- Adjustable bed for easier transfers in and out.
- Leave clear circulation space around the bed in case a wheelchair or mobility aid needs to move around it.
- Use lightweight or movable furniture so the layout can be adjusted without major effort.

**Design implication:**
Furniture placement should prioritize clear movement and accessibility.

**Reduced Physical Strain in Daily Tasks**

MS often causes fatigue, which means everyday activities can become exhausting.
Future planning considerations:

- Design layouts that reduce unnecessary movement between work zones.
- Place appliances and storage where they can be accessed with minimal effort.
- Ensure surfaces and fixtures require minimal strength or grip to use.

Examples:

- Lever handles instead of knobs.
- Pull-out drawers instead of deep cupboards.
- Easy-to-reach controls.

**Design implication:**
The home should allow Lucy to complete tasks with the least amount of physical effort possible.

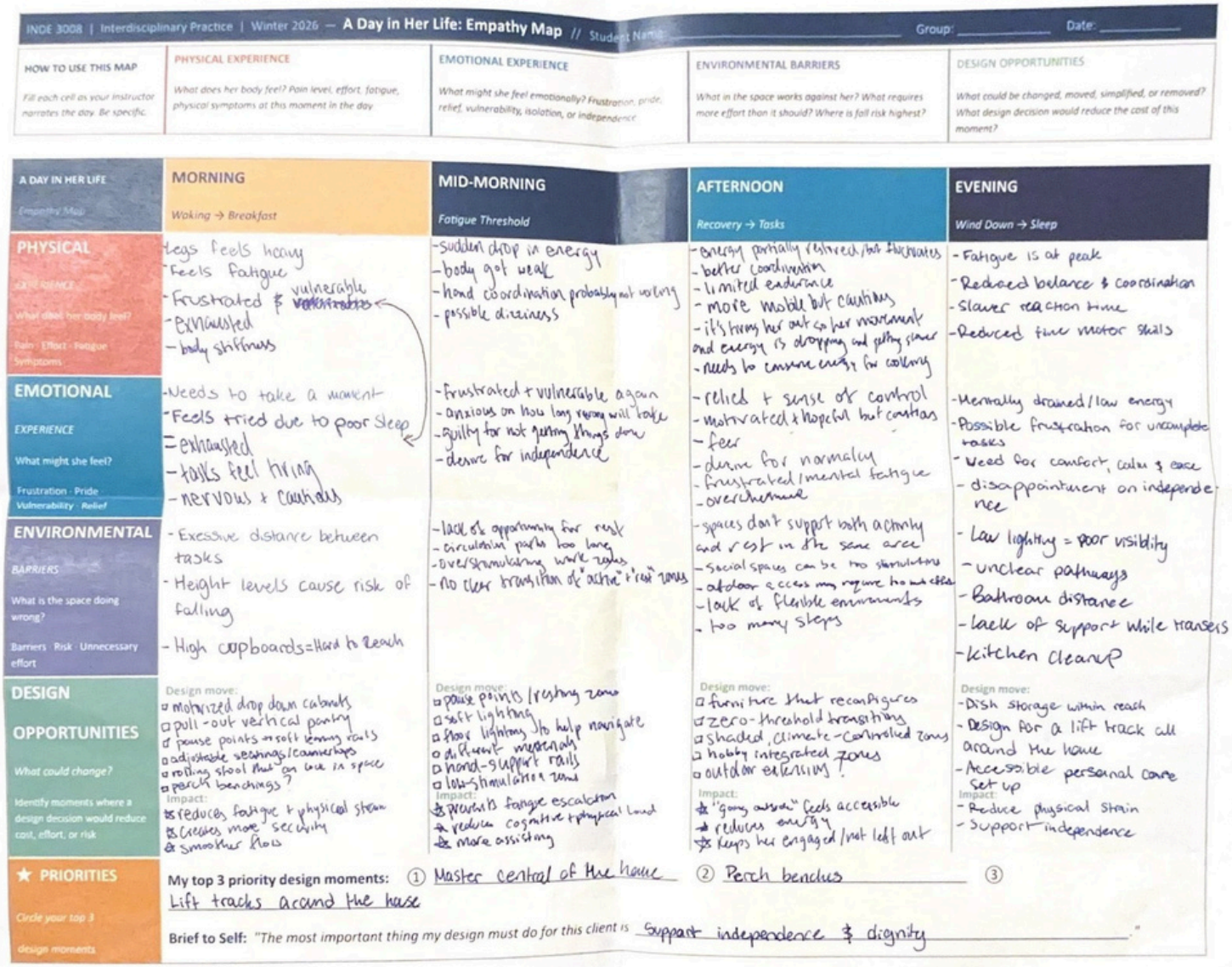
**Adaptability Without Major Renovations**

One of the biggest goals of future planning is avoiding expensive renovations later.
That means designing with things like:

- Flexible layouts
- Accessible clearances
- Universal design principles
- Adjustable fixtures where possible

**Design implication:**
The apartment should evolve with Lucy's needs instead of needing to be redesigned.

Appendix - 2nd Milestone



INDE 3008 | Interdisciplinary Practice | Winter 2026 — **A Day in Her Life: Empathy Map** // Student Name: _____ Group: _____ Date: _____

| HOW TO USE THIS MAP | PHYSICAL EXPERIENCE | EMOTIONAL EXPERIENCE | ENVIRONMENTAL BARRIERS | DESIGN OPPORTUNITIES |
|---|---|---|---|---|
| *Fill each cell as your instructor narrates the day. Be specific.* | *What does her body feel? Pain level, effort, fatigue, physical symptoms at this moment in the day* | *What might she feel emotionally? Frustration, pride, relief, vulnerability, isolation, or independence* | *What in the space works against her? What requires more effort than it should? Where is fall risk highest?* | *What could be changed, moved, simplified, or removed? What design decision would reduce the cost of this moment?* |

| A DAY IN HER LIFE — Empathy Map | MORNING — Waking → Breakfast | MID-MORNING — Fatigue Threshold | AFTERNOON — Recovery → Tasks | EVENING — Wind Down → Sleep |
|---|---|---|---|---|
| **PHYSICAL** EXPERIENCE — What does her body feel? — Pain · Effort · Fatigue · Symptoms | - Legs feels heavy<br>- Feels fatigue<br>- Frustrated & vulnerable<br>- Exhausted<br>- body stiffness | - sudden drop in energy<br>- body got weak<br>- hand coordination probably not working<br>- possible dizziness | - energy partially restored but decreases<br>- better coordination<br>- limited endurance<br>- more mobile but cautious<br>- it's tiring her out to her movement and energy is dropping and getting slower<br>- needs to ensure energy for cooking | - Fatigue is at peak<br>- Reduced balance & coordination<br>- Slower reaction time<br>- Reduced fine motor skills |
| **EMOTIONAL** EXPERIENCE — What might she feel? — Frustration · Pride · Vulnerability · Relief | - Needs to take a moment<br>- Feels tried due to poor sleep<br>- exhausted<br>- feels tiring<br>- nervous & cautious | - Frustrated + vulnerable again<br>- anxious on how long recovery will take<br>- guilty for not getting things done<br>- desire for independence | - relief & sense of control<br>- motivated & hopeful but cautious<br>- fear<br>- desire for normalcy<br>- frustrated / mental fatigue<br>- overwhelmed | - Mentally drained / low energy<br>- Possible frustration for uncomplete tasks<br>- Need for comfort, calm & ease<br>- disappointment on independence |
| **ENVIRONMENTAL** BARRIERS — What is the space doing wrong? — Barriers · Risk · Unnecessary effort | - Excessive distance between tasks<br>- Height levels cause risk of falling<br>- High cupboards=Hard to reach | - lack of opportunity for rest<br>- circulation paths too long<br>- overstimulating work zones<br>- no clear transition of "active" "rest" zones | - spaces don't support both activity and rest in the same area<br>- social spaces can be too stimulating<br>- outdoor access may require too much effort<br>- lack of flexible environments<br>- too many steps | - Low lighting = poor visibility<br>- unclear pathways<br>- Bathroom distance<br>- Lack of support while transfers<br>- kitchen cleanup |
| **DESIGN OPPORTUNITIES** — What could change? — Identify moments where a design decision would reduce cost, effort, or risk | Design move:<br>☐ motorized drop down cabinets<br>☐ pull-out vertical pantry<br>☐ pause points = soft leaning rails<br>☐ adjustable seatings/counterstops<br>☐ rolling stool that can use in space<br>☐ perch benchings?<br>Impact:<br>☆ reduces fatigue + physical strain<br>☆ creates more security<br>☆ smoother flow | Design move:<br>☐ pause points / resting zone<br>☐ soft lighting<br>☐ floor lighting to help navigate<br>☐ different materials<br>☐ hand-support rails<br>☐ low-stimulation zone<br>Impact:<br>☆ prevents fatigue escalation<br>☆ reduce cognitive + physical load<br>☆ more assisting | Design move:<br>☐ furniture that reconfigures<br>☐ zero-threshold transitions<br>☐ shaded, climate-controlled zones<br>☐ hobby integrated zones<br>☐ outdoor extensions<br>Impact:<br>☆ "going outside" feels accessible<br>☆ reduces energy<br>☆ keeps her engaged / not left out | Design move:<br>- Dish storage within reach<br>- Design for a lift track all around the house<br>- Accessible personal care set up<br>Impact:<br>- Reduce physical strain<br>- Support independence |
| **★ PRIORITIES** — Circle your top 3 design moments | **My top 3 priority design moments:** ① Master control of the house ② Perch benches ③ Lift tracks around the house | | | |

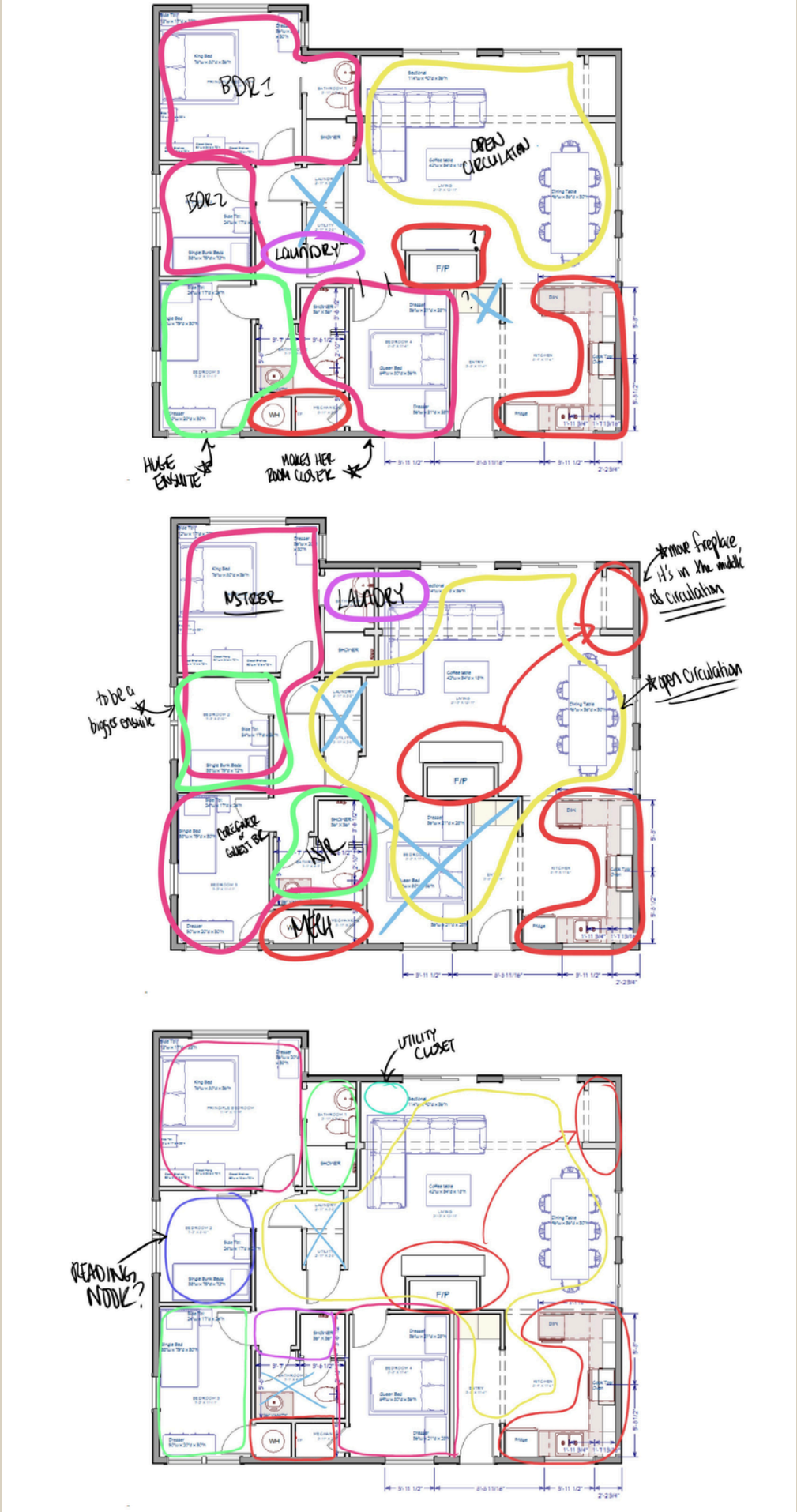
**Brief to Self:** *"The most important thing my design must do for this client is* Support independence & dignity *"*

# Appendix - 3rd Milestone

# Appendix - 3rd Milestone



# Appendix - 3rd Milestone





# Living Without Barriers

Designing an Accessible Cottage for Living with Multiple Sclerosis

## THANK YOU